

Project Design Phase Solution Architecture

Date	13 July 2026
Team ID	Bookstore Team
Project Name	Bookstore - Online Book Store Management System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

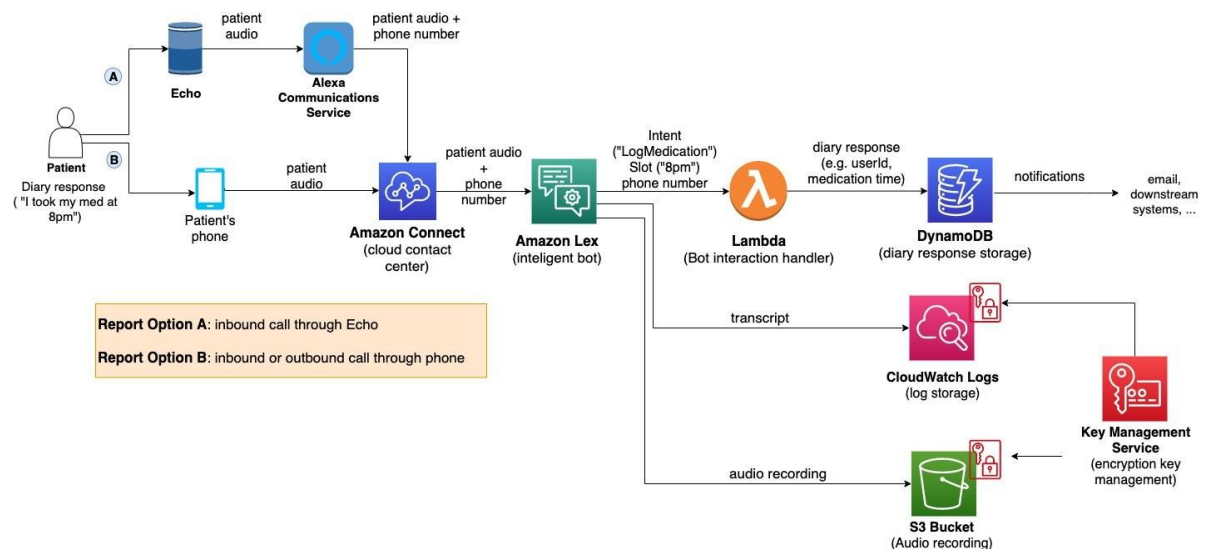


Figure 1: Bookstore Solution Architecture

Reference: Project-specific architecture for Bookstore.

Bookstore Architecture Overview

Bookstore is a production-ready online bookstore built using the MERN stack following MVC architecture. React.js provides the user interface, Node.js and Express.js expose REST APIs, MongoDB stores application data, and JWT secures authentication. The system supports three roles: Customer, Seller, and Admin.

Components:

- React Frontend
- API Gateway / Express Server
- Authentication Service
- Book Service
- Cart Service
- Order Service
- Review Service
- Admin & Seller Services
- MongoDB Database
- Multer for image uploads